StudyFlow - Student Planner

Week 1: Problem Project Proposal

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1 Problem statement and context

Software Engineering 2 students must balance class times, concurrent assessment deadlines and independent study. A deadline list alone does not connect due dates with remaining work and available study time, making it difficult to decide what to work on next and avoid overplanning. This project focuses on the five Group 1 students at CTU Training Solutions. We will document our current planning methods and concrete examples to validate these needs before finalising the requirements.

2 Target stakeholders and client

Client and primary users: the five Group 1 Software Engineering 2 students listed above. We are building the planner for our own use and will provide planning scenarios, prioritise requirements and review prototypes. Other willing students may provide additional usability feedback. **Project supervisor:** Jabulile Mashibini reviews and approves the project and assesses the work.

3 Proposed solution and initial scope

Aim: design, develop and evaluate a responsive planner that helps our group organise academic commitments and make clearer short-term study decisions.

Each student will have a private account, subjects and weekly classes, assignments/tests/exams with deadlines and remaining-effort estimates, and dated study sessions in a weekly view. A dashboard will explain which tasks are most urgent and warn when planned study exceeds the student's daily budget or overlaps a class/session. Task status and completion counts will support progress review.

Scope: one student-facing web application; no lecturer portal, shared editing, file uploads, external calendar integration or background notifications. **Proposed stack:** React, TypeScript and Tailwind CSS (Vite), with Firebase Authentication and Cloud Firestore. The data-design approach will be confirmed with the supervisor against the database deliverables.

4 Proposed high-level SE methodology

Approach: short iterative development cycles, allowing requirements and usability feedback to improve each working version within the course schedule.

Discover and design: record student needs, compare at least three existing planners, prioritise testable requirements, and produce the SRS, architecture, data model and wireframes.

Build and verify: divide features among all five members; use Git branches, meaningful individual commits and peer review. Test validation, account isolation, deadline/priority rules and study-budget calculations as features are integrated.

Evaluate and refine: run realistic planning tasks, record results and defects, gather feedback and retest fixes. Acknowledge bias when developers also evaluate the app. Use synthetic demonstration data and obtain consent for additional participants. Record supervisor approval before proceeding with the full dissertation.